Product Requirements Document (PRD) - Fintech360

🚨 The Problem: When Fintech Data Goes Out of Control

In a growing fintech, data volumes explode fast. Millions of daily micro-transactions, loan state changes, and external API calls can quickly create data chaos. If a data engineer doesn't manage this scale properly, the entire system chokes—credit risk models get fed bad data, BI dashboards take minutes to load, and financial reporting becomes untrustworthy.

Fintech360 was built to demonstrate how to tame this chaos.

This project showcases a production-ready blueprint for handling massive data growth. It proves that no matter how huge or unruly the incoming data gets, implementing strict schema validation, quarantine routing, and decoupled distributed compute (PySpark/Databricks) keeps the enterprise data ecosystem fast, accurate, and unbreakable.

1. Background & Problem Statement

Currently, credit risk decisioning and loan recovery tracking rely on fragmented data and slow reporting infrastructure. Queries take too long to execute, and risk models aren't getting the freshest data. We need a unified, scalable pipeline that feeds real-time risk features to the underwriting team and speeds up BI reporting.

2. Objectives & Goals

- Centralize all loan lifecycle data (acquisition, underwriting, repayment, collections) into a single source of truth.
- Build automated pipelines for real-time risk feature generation.
- Drastically cut down report generation time and query latency.
- Ensure data governance and PII security per RBI regulatory standards.

3. Key Success Metrics (KPIs)

- **Performance:** Reduce report generation time by 70% and improve query execution by 40%.
- **Data Freshness:** Move from batch (T+1) to near real-time for critical risk and collections features.
- **Business Impact:** Faster identification of early DPD (Days Past Due) accounts, better roll-rate tracking, and more accurate cohort analysis for the collections team.

4. Scope

- **In Scope:** AWS RDS extraction, Databricks Delta Lake setup (Medallion architecture), PySpark transformations, and Power BI semantic model optimization.
- **Out of Scope:** Core loan origination system (LOS) backend code changes, direct machine learning model deployment (this project strictly provides the *features* for the models).